WORKED SOLUTION

STEP I 2010, Question 3

Pure • Trigonometry

> **STEP I 2010, Question 3**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/10-s1-q3

The key idea

The sum-to-product formulae are what turn a difference of sines (or cosines) into a product. A chord of the ellipse joining the points with parameters u and v has gradient $\dfrac{b(\sin v - \sin u)}{a(\cos v - \cos u)}$. Both the top and the bottom are differences, so the formulae factor them, a common factor cancels, and the gradient depends only on the *sum* $u + v$. Comparing two chords then comes down to comparing two cotangents.

Solution

The trigonometric identities

Using the compound-angle formulae,

$$\begin{aligned}\sin(x+y) - \sin(x-y) &= (\sin x \cos y + \cos x \sin y) - (\sin x \cos y - \cos x \sin y)\\ &= 2\cos x \sin y,\end{aligned}$$

as required. Now, given any A and B, put $x = \frac{1}{2}(A+B)$ and $y = \frac{1}{2}(A-B)$, so that $x + y = A$ and $x - y = B$. The identity becomes

$$\sin A - \sin B = 2\cos \tfrac{1}{2}(A+B)\ \sin \tfrac{1}{2}(A-B).$$

In the same way,

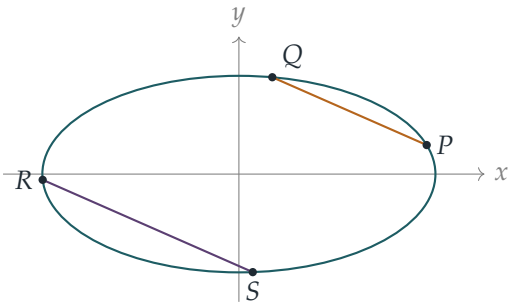
$$\begin{aligned}\cos(x+y) - \cos(x-y) &= (\cos x \cos y - \sin x \sin y) - (\cos x \cos y + \sin x \sin y)\\ &= -2\sin x \sin y,\end{aligned}$$

and the same substitution gives

$$\cos A - \cos B = -2\sin \tfrac{1}{2}(A+B)\ \sin \tfrac{1}{2}(A-B).$$

The parallel chords

All four points lie on the ellipse $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$, taken anticlockwise in the order P, Q, R, S. The figure shows a parallel pair with $a=2$, $b=1$.



Gradient of a chord. Take two parameters $u<v$ in $[0,2\pi)$ such that the chord through $(a\cos u, b\sin u)$ and $(a\cos v, b\sin v)$ is not vertical, that is, $\cos v \neq \cos u$. By the two identities with $A=v$, $B=u$, its gradient is

$$\frac{b(\sin v-\sin u)}{a(\cos v-\cos u)}=\frac{2b\cos\frac{1}{2}(u+v)\,\sin\frac{1}{2}(v-u)}{-2a\sin\frac{1}{2}(u+v)\,\sin\frac{1}{2}(v-u)}=-\frac{b}{a}\cot\tfrac{1}{2}(u+v).$$

The cancellation is legitimate: $0<v-u<2\pi$, so $\frac{1}{2}(v-u)$ lies strictly between 0 and π and $\sin\frac{1}{2}(v-u)\neq 0$. Since the original denominator is non-zero, $\sin\frac{1}{2}(u+v)\neq 0$ too, so the cotangent is defined.

Write $\alpha=\frac{1}{2}(p+q)$ and $\beta=\frac{1}{2}(r+s)$. Neither PQ nor SR is vertical, so

$$\text{gradient of } PQ=-\frac{b}{a}\cot\alpha, \qquad \text{gradient of } SR=-\frac{b}{a}\cot\beta,$$

with $\sin\alpha\neq 0$ and $\sin\beta\neq 0$. Two non-vertical lines are parallel exactly when their gradients are equal, and since $b/a\neq 0$,

$$PQ\parallel SR \iff \cot\alpha=\cot\beta.$$

? **Ask yourself** Why can't we just "cancel the cot" and conclude $\alpha=\beta$?

Because cot is not one-to-one: it repeats with period π. To keep track of this, write the difference as a single sine:

$$\cot\alpha-\cot\beta=\frac{\cos\alpha\sin\beta-\sin\alpha\cos\beta}{\sin\alpha\sin\beta}=\frac{\sin(\beta-\alpha)}{\sin\alpha\sin\beta}.$$

The denominator is non-zero, so

$$PQ\parallel SR \iff \sin(\beta-\alpha)=0 \iff \beta-\alpha=k\pi \text{ for some integer } k.$$

Pinning down k. We have

$$\beta - \alpha = \tfrac{1}{2}[(r-p)+(s-q)].$$

Since $p < r$ and $q < s$, both brackets are positive, so $\beta - \alpha > 0$. Since all four parameters lie in $[0, 2\pi)$, each bracket is less than 2π, so $\beta - \alpha < 2\pi$. The only multiple of π strictly between 0 and 2π is π itself, so $k = 1$. Therefore

$$PQ \parallel SR \iff \tfrac{1}{2}(r+s-p-q) = \pi \iff r+s-p-q = 2\pi.$$

Every step is an equivalence, so this proves both directions, as required.

(The two lines can never coincide: a straight line meets an ellipse in at most two points, and P, Q, R, S are four distinct points. So "parallel" here means genuinely parallel and distinct.)

Checks

- Take $p = 0$, $q = \frac{\pi}{2}$, $r = \pi$, $s = \frac{3\pi}{2}$, so that $r+s-p-q = 2\pi$. Then $P = (a, 0)$, $Q = (0, b)$, $R = (-a, 0)$, $S = (0, -b)$, and both PQ and SR have gradient $-b/a$. With $a = 2$, $b = 1$ both gradients come out as -0.5.
- Symmetry: the condition says $\frac{1}{2}(r+s) = \frac{1}{2}(p+q) + \pi$. The point $(x, y) \mapsto (-x, -y)$ adds π to a parameter, and rotating the chord PQ through a half-turn gives a parallel chord, which fits.
- Numerically, with random a, b and random $p < q < r < s$, the gradients agreed exactly in the cases with $r+s-p-q = 2\pi$ (about 5000 constructed examples) and never otherwise.

Common slips

- Going from $\cot\alpha = \cot\beta$ straight to $\alpha = \beta$. That gives $r+s = p+q$, which is impossible here. The period of cot is the whole point.
- Forgetting to say why $\sin\frac{1}{2}(v-u)$ can be cancelled, or not using the ordering $p < q < r < s$ to rule out $k = 0$ and $k = 2$.

Beyond STEP

Writing the ellipse as $(a\cos t, b\sin t)$ shows it as the image of the unit circle under the stretch $(x, y) \mapsto (ax, by)$. A linear map like this sends parallel lines to parallel lines, so the question is really about the circle. There, the chord from angle p to angle q is perpendicular to the radius at angle $\frac{1}{2}(p+q)$, and the result becomes almost obvious. Ideas like this one, where properties preserved by a map are used to simplify a problem, run through linear algebra and geometry. Strang's *Introduction to Linear Algebra* is a good place to see linear maps at work. Siklos's *Advanced Problems in Mathematics* has more STEP-style practice with trigonometric identities.

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